[I]NSTANT

[DAY 4]

SAFETY, GUARDRAILS & INTERNAL EVALUATION

Prepared by the Instant Day 4 SME Council

THE COUNCIL BEHIND THIS SESSION

Designed by Three Specialists, Delivered as One Session

[01]

Responsible AI & Clinical Safety Officer

Lead Trainer — Day 4

Owns confidence thresholds, unsupported claim detection, and clinical risk.

[02]

Evaluation & Metrics Specialist

Content Advisor

Defines how Precision@k, citation accuracy, and faithfulness are scored.

[03]

Human Factors & Automation Bias Advisor

Technical Advisor

Ensures uncertainty is communicated honestly to end users.

LEARNING OBJECTIVES

By the End, Every Team Can:

1. Set and justify a retrieval confidence threshold
2. Detect when a generated answer contains an unsupported claim
3. Compute Retrieval Precision@k, citation accuracy, and faithfulness
4. Explain automation bias and why overconfident AI is dangerous in care settings
5. Write uncertainty language that is honest without being useless
6. Run a full internal evaluation suite and interpret the results

MODULE 0

Recap & Today's Goal

From cited answers to measurably safe answers

What You Walked In With

A grounded prompt

Stress-tested against adversarial questions

A cited answer format

Recommendation, excerpt, and citation every time

A working refusal case

Rehearsed and ready for the Day 5 demo

What's missing: proof, in numbers, that it's safe to trust.

Confident Isn't the Same as Correct

A system that always sounds sure of itself is more dangerous than one that says “I'm not certain.” Today you build the guardrails and the numbers that prove your system knows the difference.

01
Confidence Thresholds

→

02
Evaluation Metrics

→

03
Responsible AI & Safety

→

04
Run the Full Eval Suite

MODULE 1

Confidence Thresholds & Claim Detection

Teaching your system to know what it doesn't know

A Gate Before Generation Happens

A confidence threshold is a minimum similarity score your retrieved chunks must clear before the model is allowed to generate an answer at all. Below the threshold, refusal takes over automatically.

- Retrieve top-k chunks
- Check top similarity score
- Score ≥ threshold → generate
- Score < threshold → refuse

[Starting Point]

Threshold ≈ 0.65–0.75

Use your Day 2 Precision@k logs to calibrate — the score where relevant chunks reliably outrank irrelevant ones is your threshold.

Every Threshold Is a Trade-off

Threshold Too Low

- Weak matches still trigger generation
- Higher risk of answering from irrelevant context
- More hallucination risk

Threshold Too High

- System refuses questions it could actually answer
- Frustrates users with unnecessary refusals
- Looks less capable than it is

Catching What Slips Past the Prompt

Even a well-grounded prompt can occasionally drift. A second, independent check compares the generated answer against the retrieved text — a safety net, not a substitute for good prompting.

1

Split

Break the generated answer into individual factual claims

2

Match

Check whether each claim appears in the retrieved passages

3

Flag

Any claim without a matching source is flagged as unsupported

MODULE 2

Evaluation Metrics

The three numbers your judges will ask for by name

Three Metrics That Matter Most

Retrieval Precision@k

Are the retrieved chunks actually relevant?

BUILT ON YOUR DAY 2 TEST SET

Citation Accuracy

Does the citation match a real, correct source location?

NEW FOR TODAY

Faithfulness

Does the answer contain only facts present in the retrieved text?

NEW FOR TODAY

Scoring Citation Accuracy

Citation Accuracy = (correct citations) ÷ (total citations given)

A citation is “correct” only if all three hold:

- ✓ The document name actually exists in your source library
- ✓ The section and page number are accurate
- ✓ The cited text genuinely supports the claim it's attached to

Scoring Faithfulness

Faithfulness = (claims supported by retrieved text) ÷ (total claims made)

This is the metric most directly tied to hallucination. A faithfulness score below 1.0 means your system said something the evidence didn't actually support — even if the citation format looked perfect.

Target for today: 0.9+ faithfulness across your full test set.

Your Evaluation Log Template

Query	Precision@5	Citation Acc.	Faithfulness
Screening interval for breast cancer	0.80	1.00	1.00
First-line hypertension treatment	0.60	0.75	0.80
Out-of-scope question (refusal test)	N/A	N/A	N/A — refused correctly

Sample data for illustration — average these across your whole test set for the numbers you'll present Friday.

MODULE 3

Responsible AI & Clinical Safety

Building a system worthy of the trust it asks for

The Danger of Being Too Convincing

Automation bias is the tendency to trust a system's output simply because it's confident, fast, and well-formatted — even when it's wrong. In clinical settings, a fluent wrong answer can be more dangerous than an obviously broken one.

[ ]

**Your job isn't to make the system sound smart.
It's to make its confidence match its evidence.**

Calibrate the Words, Not Just the Numbers

Evidence Strength	Language to Use
Strong, direct match	*“The guideline recommends…”*
Partial or indirect match	*“The guideline suggests, though this doesn't directly address…”*
Weak or conflicting match	*“Limited evidence found; consider consulting the full guideline…”*
Below threshold	*Refuse — do not soften into a guess*

Responsible AI Checklist for Clinical Context

1. No answer implies it replaces clinical judgment
2. Uncertainty language matches actual evidence strength
3. Refusals never get quietly softened for the sake of a demo
4. A disclaimer is visible, not buried in fine print

MODULE 4

Internal Evaluation Exercise

Run the numbers on your own system, right now

From Numbers to Next Steps

1. Run your Day 2 test set through the full pipeline (retrieval + generation + citation)
2. Score every result for Precision@k, citation accuracy, and faithfulness
3. If faithfulness is low: check your grounding prompt before touching retrieval
4. If citation accuracy is low: check your metadata schema before touching the model
5. Log every score — this is what backs up tomorrow's demo

Day 4 Competition Checklist

- ✓ Confidence threshold set and calibrated against Day 2 data
- ✓ Uncertainty language calibrated to evidence strength
- ✓ Unsupported claim detection implemented and tested
- ✓ Responsible AI checklist reviewed as a team
- ✓ Precision@k, citation accuracy, faithfulness all computed
- ✓ Full evaluation log saved and ready to present

Common Day 4 Pitfalls to Avoid

One threshold for every query type

Not accounting for how query difficulty varies across topics

Metrics computed once, never rerun

Not re-scoring after fixing an issue, so the fix is never verified

Confident language regardless of score

The prompt still sounds certain even when faithfulness is low

Skipping the team review

Nobody actually reads the responsible AI checklist together

What “Done” Looks Like at the End

1. A calibrated confidence threshold with unsupported-claim detection running live
2. Full evaluation numbers: Precision@k, citation accuracy, and faithfulness
3. Uncertainty language that honestly reflects evidence strength
4. A team that can defend every safety decision in front of the judges tomorrow

Trust, Backed by Numbers

Questions? Find your Module trainer before you head to your team table.